



EDUCATION

San Diego State University

Bachelor of Science in Computer Engineering

Expected Graduation: May 2026

Relevant Coursework: Machine Learning, Circuit Analysis 1 & 2, Data Structures, Digital Systems, Computer Organization, Linear Algebra, Discrete Math, Microprocessors, Database and Web Programming, Signals and Systems, Digital Circuits, Embedded System Programming

EMPLOYMENT

Sparkl Reusables - Contract Software Developer

May 2023 - Present

- Implemented the full RFID lifecycle on mobile, from initial tag assignment and bulk registration to tracking incoming returns and outgoing tags
- Developed a real-time synchronization engine between a handheld scanner and a PHP-based backend API, ensuring field-level data consistency with a live SQL database.
- Built a responsive, reactive UI using Jetpack Compose to manage high-volume RFID queues, featuring real-time validation of scanned tags against specific product categories (inventory_objects).

PROJECTS

Quetzal Drone

September 2025 - Present

- Led a team of 5 electrical and avionics integration for a VTOL competition drone as ECE Lead.
- Integrated Pixhawk 6X, GPS, telemetry, and ESC outputs; verified actuator mapping and system communication in QGroundControl.
- Performed system integration testing across power, motors, telemetry, hover, and transition flight phases.
- Led the autonomy development on Jetson Orin Nano using a YOLO-based vision pipeline for target detection with a 7,000+ image dataset and ~410 ms inference latency
- Solely developed quetzal-drone.com, a project website showcasing the drone's design, testing, and competition progress

Release IOS Application

- Built a cross-platform student subleasing app (React Native + Expo + TypeScript) with a production-ready mobile/web codebase and shared component architecture.
- Designed an AI-ready listings data pipeline (Supabase + local SQLite caching) to support future ranking, recommendation, and personalization workflows.
- Shipped with modern mobile tooling (EAS build/release, Expo Router, typed APIs), enabling fast iteration on AI-powered product experiments like smart matching and automated listing insights.

Super Vape Bros Website

December 2025 - Present

- Developed responsive front-end functionality using HTML, CSS, and JavaScript with an integrated customer-facing promotional feature into an existing website workflow.
- Implemented reusable UI components, animations, and event-driven user interactions.
- Iterated on design and functionality based on testing and stakeholder feedback.

Waymark — Collaborative Trip Planner (team of 5)

January 2025 - May 2025

- Built a web platform for shared travel itineraries with budgeting, including user profiles, trip creation/editing, destination recommendations with ratings/reviews, and real-time collaboration across frontend (TypeScript/React) and backend (Python).
- Contributed 188 commits (~37% of total project activity) as one of the top contributors in a 5-person development team.
- Demonstrated version control expertise, managing branches, merges, and feature integration in a 500+ commit repository.

FM King Inc

October 2025 - Present

- Designed and built fmkinginc.com, a responsive marketing site for a licensed construction contractor, highlighting services
- Created a before-and-after project portfolio to showcase completed remodels and renovations, improving visual storytelling and credibility for prospective customers.
- Developed a lead-capture contact form that collects name, email, phone, project type, and project details, streamlining incoming project inquiries and quote requests for the business owner.

ROLL AND RISE: The Serpentine Challenge (Game Development)

January 2025 - May 2025

- Designed a Snakes-and-Ladders-style game in MIPS assembly and C. Features include snakes, ladders, chance cards, dice rolls (including zero outcomes), a moving "ghost" player, and a bitmap display. Engineered efficient low-level input handling for interactive gameplay, integrating graphics and system calls.

SKILLS

Technical: Machine Learning, YOLO, Full-stack web development, C, C++, Python, CSS, Tailwind CSS, React, React Native, Expo, JavaScript, HTML, Docker, SQL, MySQL, PostgreSQL, VS Code, WebStorm, Soldering, Git, Github, Figma, ShadCn, JWT authentication, MIPS Assembly, Vivado, FPGAs, data structures, circuit analysis, data analysis, order management, Autodesk, AutoCAD Revit, AutoCAD Inventor, Solidworks, surveying, math, physics, Excel, Word, PowerPoint, Gantt Charting, Data Tracking & Reporting, Email Marketing (HTML)